BUSINESS PROPOSAL

SynaptOS

Retail Nervous System

Agentic AI that eliminates fresh-food waste and maximises perishable revenue - autonomously.

$15.2B	17.2%	$1.5M	90%+
Global TAM by 2034	Market CAGR	Seed Round Target	Gross Margin

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1. EXECUTIVE SUMMARY

SynaptOS is an Agentic AI operating system that is meant to serve fresh-food retail supply chains. It connects to the existing POS support and executes the entire price determination procedure of perishable stock: reads real-time inventory amounts and expiry dates, and decides to alter prices to virtual shelf tags and self-optimizes itself on the premise of the later selling rate.

Its underlying remedy is a commercially acute problem, Vietnamese grocery retailers enjoy 2-3 net margin, and waste 15-25 of their fresh commodities to rot. The existing POS systems are mere books - they record the buys but contain no information of the expiry date, weather variations in demand and the purchasing power of the neighbourhood. Every decision to be made on pricing is a manual gut-call, too late and too crude.

That gut-call is replaced by SynaptOS with a fixed AI reasoner loop that costs $0.015 a cycle and yields 5-50 dollars of perishable value / decision. The arbitrage gross margin is more than 90.

> The vison thesis
>
> SynaptOS converts the biggest cost in the grocery business, spoilage, into the biggest income line, agentic AI that no incumbent or global competitor has created in Southeast Asia.

Key metrics

$15.2B	17.2%	$0.015	333×
Global TAM by 2034	Market CAGR	Cost per AI loop	Min ROI per loop

$75	5%	8-16×	SDG 12
SaaS / store / month	Efficiency fee on rescued GMV	LTV : CAC ratio	UN Goal Alignment

2. THE PROBLEM

2.1 FRESH FOOD IS A TICKING CLOCK

The structure of economics of fresh-food retail is aggressive. Even unsold perishable products are wasted and their worth cannot be redeemed on a defined time schedule which cannot be remedied through any form of management care. A retailer can only do is to charge it and at the appropriate time, it will sell the product before it expires. Miss the markdown, too small, too late, too big, and the product is rotted or the margin is killed.

> Perishable GMV is lost to spoilage by Vietnamese fresh-food retailers at 15 – 25% /year. This is hundreds of billions of recoverable value being lost annually on margins that are below 3%. to a 100-store chain that is generating VND 2.1 billion per store per month.

2.2 POS SYSTEMS ARE PASSIVE RECORDERS

The technology stack that was employed to take care of the vast majority of Vietnamese grocery stores was developed with a single purpose in mind, the latter being the monitoring of the transactions. POS systems are not smart engines, but books. They are being imported with no idea of:

- Perishable SKU expiry dates.
- Live inventory spoilage and depletion.
- Demand varies with weather (during hot seasons, demand on drinks rises, when there is a cold front, the demand on hot food moves to the right).
- Variations in purchasing power of the neighbourhood.
- Prices of commodities that affect the behaviour of consumer substitution.
- Purchasing risk and supplier lead times.

All the pricing decisions, such as when to mark-down, how much, what products are done, are all done by a store manager, manually using experience and intuition. The result is a failed system of markdowns: intensive 30 - 60% discounts within 24- 48 hours of expiry, over which a lesser markdown would have been put in place a priori and sold the product at a higher profit. The failure to perform is not a human issue but a systemic issue. Not a single manager is able to track 500 SKUs on the flight during 8-hour shift.

2.3 VIETNAM: HIGH-GROWTH, HIGH-URGENCY

The CAGR of Vietnam grocery market is estimated at 13.6 percent to $1T retail value in 2033 that is fueled by 7%+ GDP growth and soaring middle income group. The combination of the factors of structure makes this market in a clearly acute relation of the problem of spoilage:

Factor	Vietnam-specific dynamic	Impact
Tropical climate	Average temperatures of 28 to 35^{0}C are favorable to the growth of bacteria as well as shortening the expiry dates of all fresh categories.	30 - 40% higher rate of spoilage as compared to temperate markets.
Urbanisation rate	Due to the high population concentration in each of the HCMC and Hanoi, the cities boast of thick but very heterogeneous areas of demand.	Hyperlocal patterns of demand are a breakdown of standard forecasting models.
Chain growth	Bach Hoa Xanh 2,180+ outlets; WinMart+ intensive growth; new entrants Korean and Japanese.	Scale blows the problem of spoilage out of thousands of spots at once.
Thin digital layer	The use of POS in the form of an in-store is not very prevalent; the majority of chains continue to do their counts by hand.	There is no data infrastructure to support AI, but it has to be overlaid, as opposed to substituted.
Regulatory support	Decision 222/QD-TTg (January 2025): National Circular Economy Plan will imply that as of 2035, monitoring the food waste will be done.	The audit-ready waste-reduction data demand (policy-based, rather than optional) by enterprises has been changed.

3. THE SOLUTION - SYNAPTOS

3.1. SYSTEM OVERVIEW & UNIT ECONOMICS

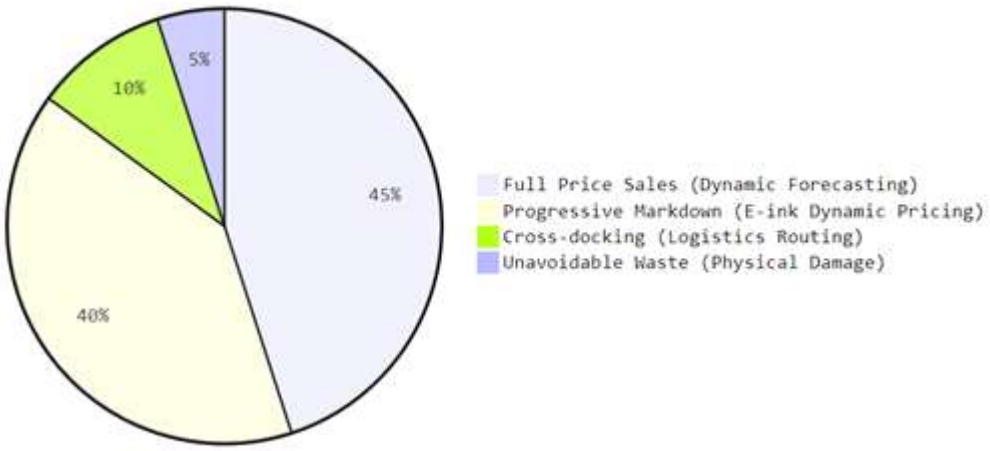
SynaptOS replaces passive, legacy POS/ERP systems with an autonomous central system for fresh-food inventory. It connects physical stock levels directly to autonomous commercial decisions.

3.1.1. Unit Economics

Instead of relying on human managers to monitor thousands of expiring SKUs, which leads to latency and margin destruction, SynaptOS executes pricing arbitrage using micro-computing power.

Core metric	Data	Impact
Cost per AI loop	$0.015	Marginal cost of inference is near zero per decision.
Value recovered	$5 - $50	Minimum 333x ROI on every successful Agent action.
Gross margin	> 90%	Driven by a $75 SaaS base + 5% Efficiency Fee on rescued goods.
Target spoilage	< 5%	Drastic reduction from the 15-25% industry average.

As illustrated in the target impact distribution below, the system relies on a multi-lever approach. It doesn't just offer discounts, it optimizes full-price sales and utilizes logistics routing to minimize unavoidable waste to 5%.

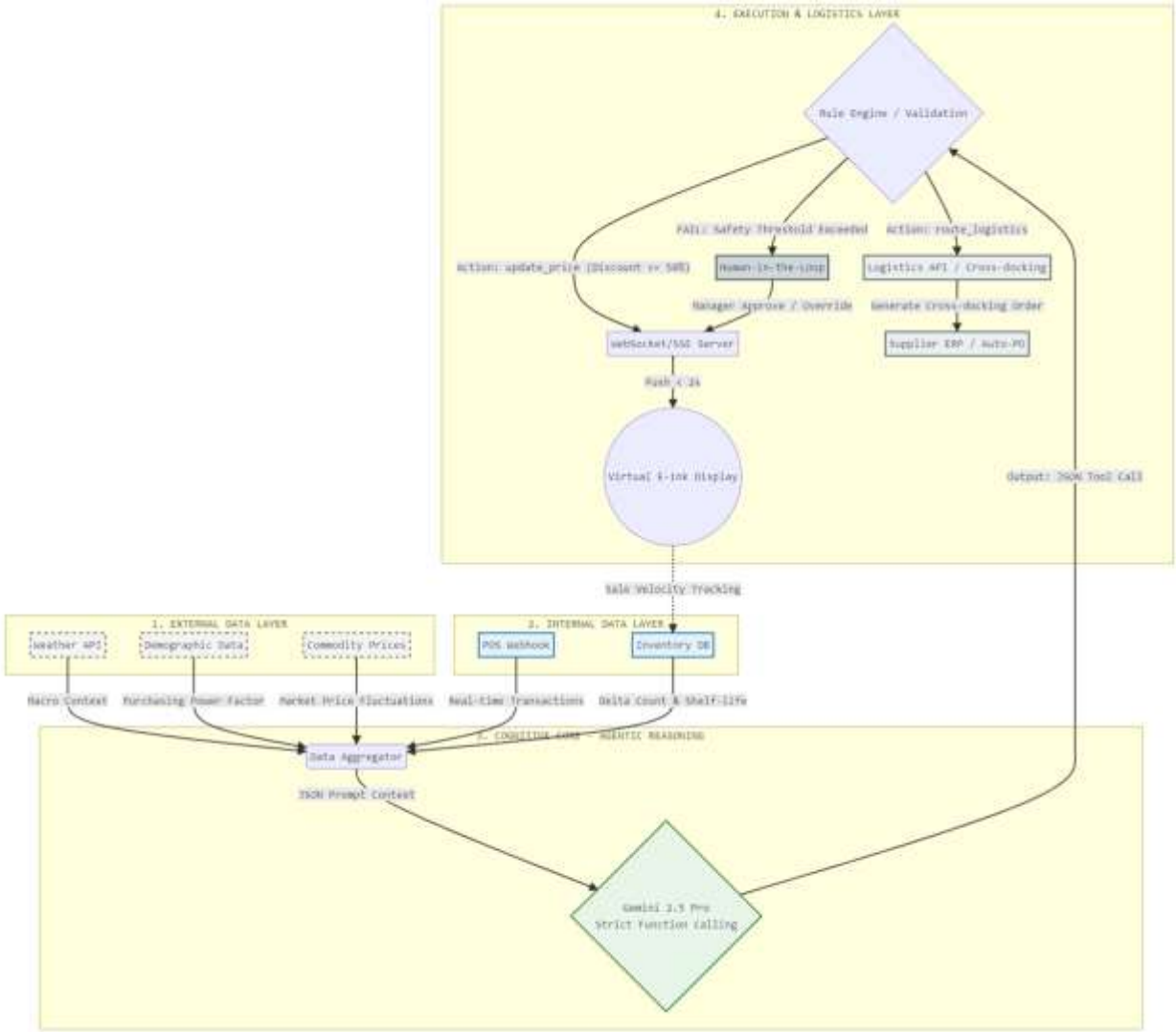


3.2. AGENTIC DATA FLOW & ARCHITECTURE

The system is decoupled into three distinct layers:

Data ingestion -> Cognitive core -> Execution layer.

The flowchart below maps the architecture technical logic: the agent aggregates both internal POS data and external macro signals to make deterministic, API-driven decisions.



Architecture breakdown:

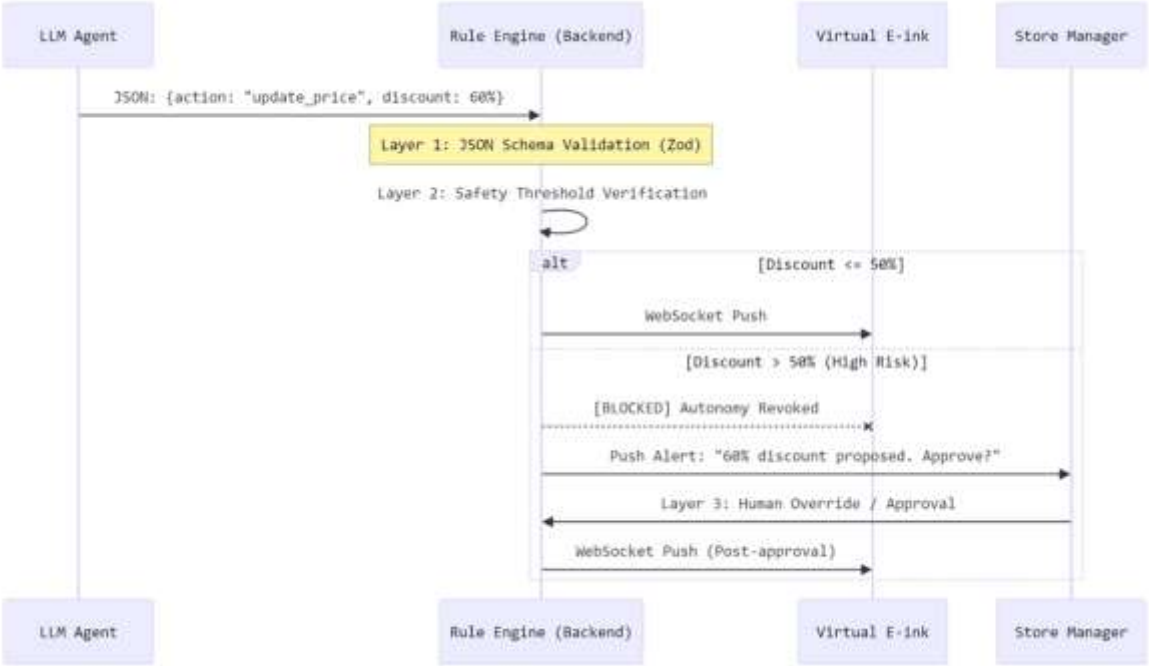
- Context assembly: The aggregator fuses internal inventory levels with external signals (e.g., heavy rain, high-income district) to form a rich reasoning context.

- Strict function calling: The LLM is prohibited from outputting free-text. It is forced to return structured JSON tool calls (either *update_price* for localized clearance, or *route_logistics* for cross-store transfer).

3.3. ANTI-HALLUCINATION GUARDRAILS

A critical failure point for AI in enterprise environments is hallucination. SynaptOS neutralizes this financial risk through a Defense-in-Depth (DiD) Middleware Architecture.

3.3.1. Core risk mitigation tactics

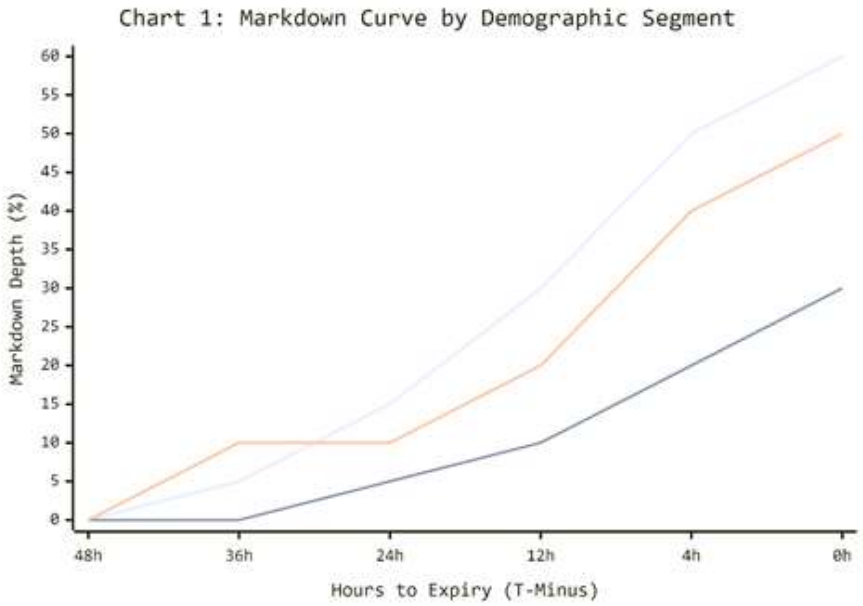


- Temperature = 0.0: The LLM operates in strict deterministic mode. "Creativity" is disabled for financial reasoning.
- Shrinkage calibration: Solves the POS blindspot. Managers must manually input physical shrinkage (stolen/damaged goods) at the end of shifts. The AI calculates markdowns based strictly on verified physical stock.

3.4. CONTEXTUAL ELASTICITY & DYNAMIC PRICING

Unlike static rule-based systems, SynaptOS segments its pricing strategy geographically, demographically, and temporally. This multi-dimensional elasticity model creates the system's primary Data Moat. The agent recognizes that a 20% discount on a Tuesday morning in a residential area performs fundamentally differently than the same discount on a Friday night at a walking street.

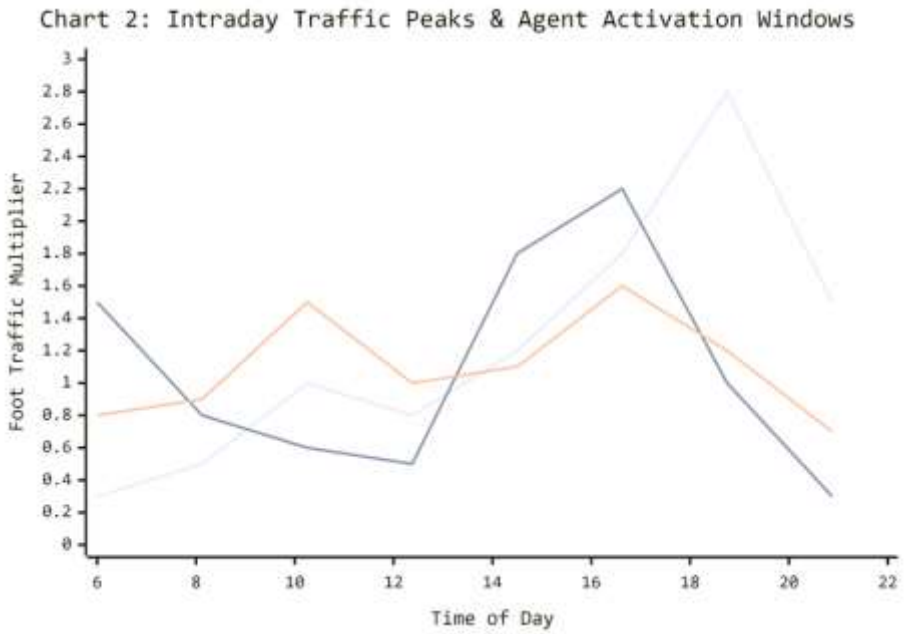
3.4.1. Demographic markdown trajectories



- Residential zones (light gray line): Prioritizes rapid capital recovery via early, aggressive markdowns.
- Premium zones (dark gray line): Defers discounts to the last possible hour to protect brand positioning.
- Transit zones (orange line): Maintains flat pricing during the day, triggering aggressive flash-sales just before foot traffic drops.

3.4.2. Intraday traffic & temporal execution

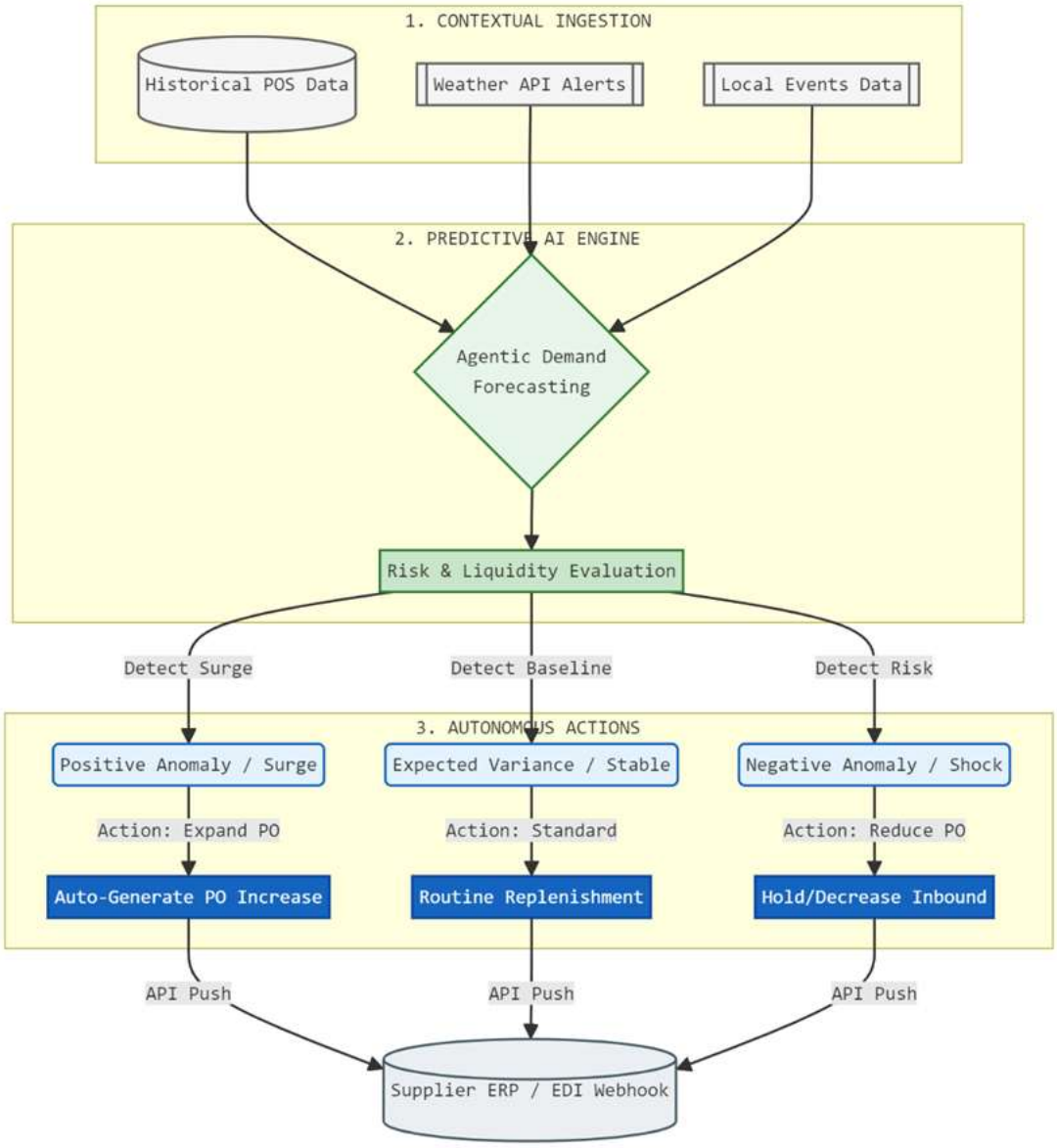
The agent maps its pricing execution to local foot-traffic spikes, ensuring markdowns are applied when liquidity is actually present.



- 08:00 (suburban morning peak): Traffic spikes as families buy daily groceries. The agent holds firm on base prices, refusing to discount early, capturing full margin from high-intent buyers.
- 12:00 (premium urban lunch rush): Office workers flood Premium zones. The agent triggers micro-markdowns (e.g., 5-10%) strictly on Ready-to-Eat (RTE) meals nearing expiry, converting perishable risk into immediate lunch sales.
- 17:00 - 18:00 (suburban post-work rush): The critical clearance window for Residential zones. The agent triggers Progressive Volume Pricing (e.g., family-pack meat discounts) to clear heavy inventory before traffic dies completely at 20:00.
- 20:00 (transit / walking street peak): While suburban stores are empty, Transit zones reach maximum liquidity. The Agent activates Aggressive Flash Sales on snacks and beverages, capturing impulse buyers and zeroing out daily inventory.

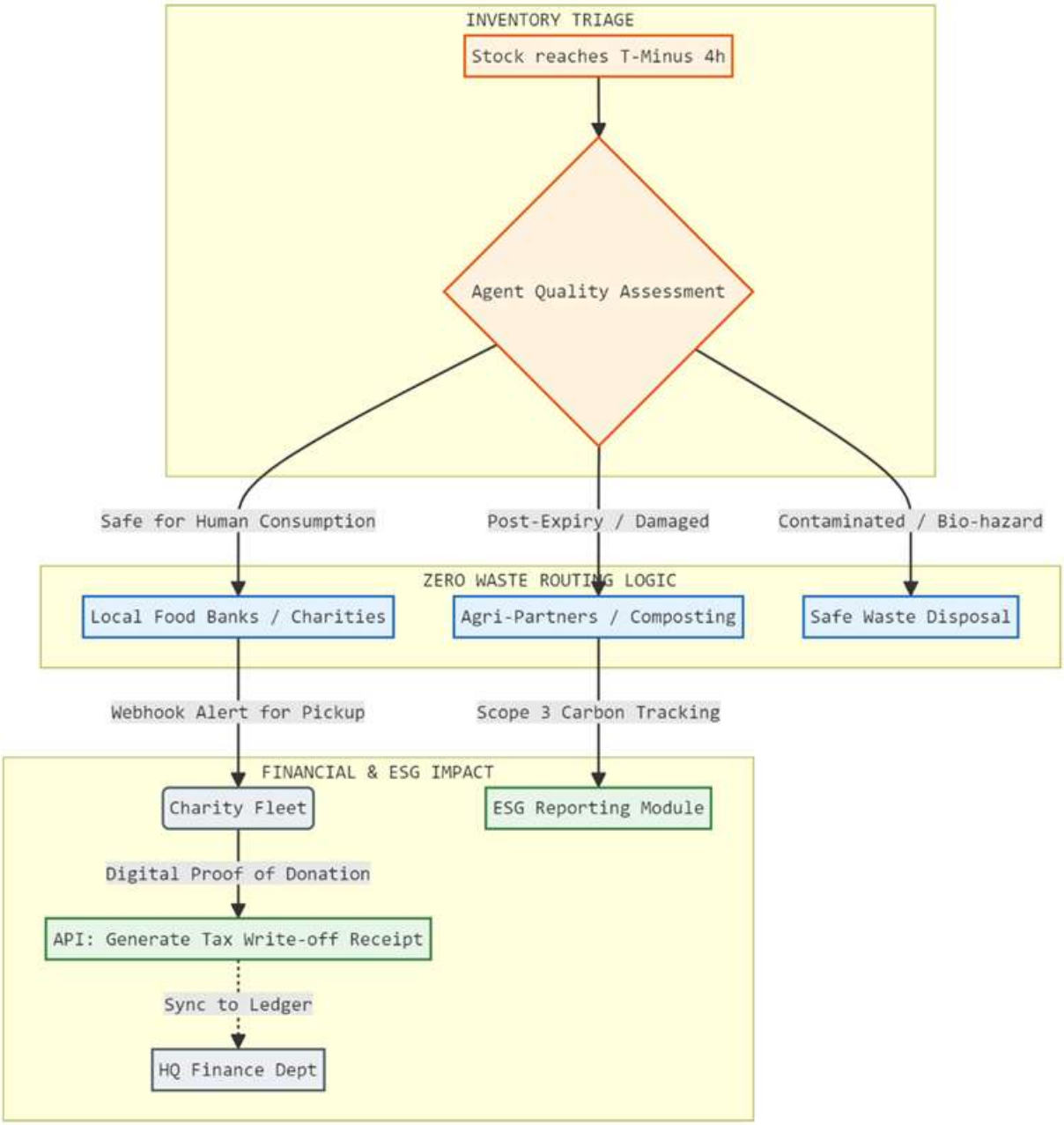
3.4.3. Predictive procurement engine (Upstream logistics)

The most effective way to eliminate waste is to prevent overstocking before it enters the store. SynaptOS utilizes a predictive procurement engine that autonomously adjusts inbound Purchase Orders (POs) based on multi-variate context.



3.4.4. End-of-Life (EOL) zero waste routing (Downstream logistics)

When commercial clearance (E-ink discounts) reaches its limit at T-Minus 4 hours to expire, the agent officially revokes the SKU from the sales floor and triggers physical EOL routing. This process converts unavoidable waste into ESG compliance and financial tax write-offs.



3.4.5. SKUs contextual strategy

The agent does not treat all expiring inventory equally. It applies hyper-localized strategies based on the "hero SKUs" (key items) of each micro-zone.

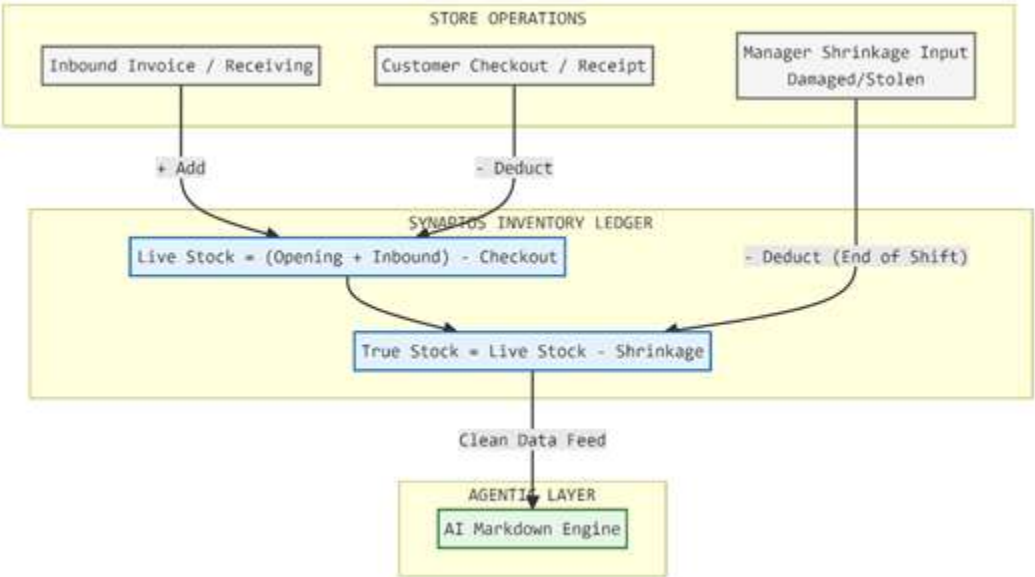
Store archetype	SKUs (High spoilage risk)	Contextual strategy
Premium urban	Organic produce, premium proteins	Cross-docking priority. The agent routes surplus to high-demand nodes rather than discounting locally, protecting margin.
Transit / walking street	Ready-to-eat (RTE), cold beverages, snacks	Weather-responsive & impulse pricing. The agent leverages immediate context (e.g., suppressing discounts on cold drinks during a 35°C heatwave) and triggers EOD flash clearance.
Suburban / residential	Family-pack meats, bulk vegetables	Progressive volume pricing. The agent applies early discounts to secure weekly family-shopper baskets before competitors.

3.5. POS OPERATIONAL ARCHITECTURE & HIERARCHY

AI cannot physically count inventory. The SynaptOS integration relies on a strict ledger-based inventory engine combined with an operational hierarchy (RBAC) to ensure data integrity.

3.5.1. The ledger & calibration loop

SynaptOS calculates real-time inventory purely through invoice and receipt of parsing, corrected by human intervention.



3.5.2. Operational hierarchy (Role-based access control)

To maintain enterprise security, system intervention is strictly tiered:

Operational level	Role function	Authorization level
Tier 1: Staff / cashier	Execution	Read-Only. Processes checkouts. Cannot alter AI pricing or inventory data.
Tier 2: Store manager	Calibration & safety	Gatekeeper. Required to input physical shrinkage (EOD). Authorized to approve/reject AI markdowns exceeding the 50% safety threshold.
Tier 3: Chain HQ / admin	Strategy	God Mode. Configures global markdown thresholds, assigns store demographic profiles, and receives aggregated ESG / Tax write-off reporting.

4. MARKET OPPORTUNITY

4.1 GLOBAL MARKET SCALE

Among the quicker emerging sectors of AI, retail technology, and sustainability compliance, the food waste management sector is one. The controlled market will expand to $15.16B in CAGR of 17.2 to become 2034 what was projected to be in 2024 $3.10B of investor ESG needs and compounding unit economics of AI-driven waste cut-off.

$3.6B	$15.2B	17.2%	52.8%
Global market 2025	Global market 2034	CAGR 2025–2034	Retail & supermarket share

4.2 TAM / SAM / SOM BREAKDOWN

Market	Definition	Size	Basis
TAM	Global AI-powered food waste management for retail and food service	$15.2B by 2034	17.2% CAGR from $3.6B 2025 base (Towards FnB, 2025)
SAM	Southeast Asia fresh-food retail AI (Vietnam, Thailand, Indonesia, Malaysia, Philippines)	$850M by 2028 est.	Vietnam $269B retail economy + SEA multiplier; 20.7% Vietnam smart retail CAGR
SOM	Vietnam Year 3 revenue target: 2,000 stores at blended $90/store/month	$18M ARR by Year 4	10% penetration of Vietnam's ~20,000 modern grocery locations; conservative efficiency fee assumptions

4.3 VIETNAM MARKET SIGNALS

Vietnam is not simply a large market - it is a market at a structural inflection point where the conditions for rapid technology adoption are aligning simultaneously:

Signal	Data Point	Implication for SynaptOS
Retail economy scale	$269B total 2025; food & beverage = 45%	Core addressable segment exceeds $120B annually
Growth trajectory	13.6% retail CAGR; GDP 7%+ in 2025	Market growing faster than our expansion timeline
Smart retail adoption	Vietnam smart retail: $238M in 2024 → $1.56B by 2033 at 20.7% CAGR	Infrastructure Retail AI and already in appetite.
Chain scale	Bach Hoa Xanh: 2,180+ stores at VND 2.1B/store/month avg.	200 or more store deals in case of 10% conversion under single chain.
E-invoice mandate	252,000 businesses on e-invoice by mid-2025; formal POS adoption accelerating	Distribution channel (POS vendors) gets up automatically.
Regulatory tailwind	Decision 222/QD-TTg (Jan 2025): Circular Economy Plan mandates food waste tracking	SynaptOS enterprise demand is a policy-based demand.

5. COMPETITIVE LANDSCAPE

5.1 OUR ADVANTAGE

Varying pricing of perishables is not a new concept. Wasteless (Israel, founded 2017, raised $15M) has demonstrated that the model works in the European supermarkets. The type of demand forecasting provided by RELEX Solutions and Afresh to large chains is an AI one. Nevertheless, Vietnamese retailers are unable to purchase all these solutions and none of the available options were developed taking into consideration the specifics of Southeast Asian market.

Capability	SynaptOS	Wasteless (IL)	RELEX	Manual Markdown
Agentic execution (acts autonomously)	YES	YES	NO - dashboard only	NO
Zero hardware requirement	YES - software only	NO - requires ESL hardware	YES	YES
Macro signal ingestion	YES - live commodity feeds	NO	Partial	NO
Geo-demographic pricing	YES - district-level profiles	NO	NO	NO
SEA market-native	YES - built for Vietnam	NO - Europe/US focus	NO	N/A
POS plugin model	YES - no rip-and-replace	NO - standalone system	NO - full ERP replacement	N/A
Autonomous procurement	YES - PO generation agent	NO	YES (RELEX)	NO
Pricing model	$75/store + 5% efficiency fee	Hardware + monthly licence	$50K+/year enterprise	Staff cost only

5.2 COMPETITIVE MOAT

Not the fact that SynaptOS is based on the AI model, the foundation models are commoditising at a very high rate because it is a formal and accrual moat:

- Data networks: Adding the number of all data on the price elasticity model. The two forms of the districts make geo-demographic pricing machine ready. The competitive information would consist of 24 months of compounding training on the basis of the entering the competition within the market in Year 3.
- POS vendor lock-in: The promotion-motivated expansive distribution share and switching share of biggest POS systems of Vietnamese SME retailers such as KiotViet and MISA are both expansive. The existing software that is owned by the retailer becomes a native plug-in of SynaptOS.
- Procurement network: The greater the degree to which the autonomous procurement agent integrates with the supplier, the greater the system value increases and the greater the cost of switching to an opponent of the supplier integration.
- First-mover geography: No native of SEA competitor. It is presumed that that would take between 18-24 months to generate information and partnership value when a global players with significant resources at its disposal to finance broadcast innovations in its country of origin localises its offerings to the local markets.

5.3 WHY NOW

Three convergent forces call on this being constructed in 2026:

- By 2024, agentic AI infrastructure will be developed. The tool-use and logic capacities to construct an effective autonomous pricing agent, had not been attained to the manufacturing quality until the functional-calling capacities of Claude 3 and GPT-4o. SynaptOS could not be created in 2022 and can be made in 2026.
- One, e-invoice compulsion in Vietnam (twenty-five percent of businesses will have to comply by mid-2025) is committing organized production of POS data production is occurring in the informal trader which is a granite making raw material of AI inventory control is forming in a mass in a manner it has never been before.
- ESG and circle economy regulation (Decision 222/QD-TTg) create the institutional necessity to utilize the information related to the food waste tracking. Enterprise groceries should have a data layer that is audit-ready; SynaptOS would provide it to them as a by-product of the existence.

6. BUSINESS MODEL

6.1 REVENUE STREAMS

The model of alignment of incentives between the platform and its customers in every phase of adoption is provided at SynaptOS in three streams of B2B SaaS.

Stream	Mechanism	Unit Price	Stage
SaaS Subscription	Monthly, paid peractive store, flat, through integration partner of POS. WebSocket E-ink protocol coordination, all AI reasoning loop, dashboard, support.	$75 / store / month	Core (live at launch)
Efficiency Fee	5% on the GMV that shall have been recovered on AI-resued perishables. Delta calculated with POS Marked down based on AI sold, otherwise would have been wastes base Recovery with synaptos is an improved response to recovery.	5% of rescued GMV (~$40–70 addl. avg.)	Core
Enterprise API	Dynamic pricing engine publicly available to POS partners (KiotViet, MISA) or any individual retail company developing an in-house tooling. Monthly platform licence.	$2,000–$15,000 / month	Phase 2 (Month 12+)

6.2 GO-TO-MARKET MOTION

It sells its bottom to generate sales via channel of POS vendors and switched to bottom anchor chain offers when initial evidence was proven.

Phase	Motion	Target	Timeline
1.Seeding	HCMC friendly pilot 3-5 stores. Efficiencyonly fee- no pilot subscription fee. Base rescue and GMF Recued. Increase of data point of recovery evidence.	3–5 stores; measurable waste reduction ≥30%	Months 0–3
2.POS Channel	KiotViet and MISA veto certification as plug-in. Their clients who are retailers are	POS vendor partnership agreements	Months 3–6

Phase	Motion	Target	Timeline
	introducing SynaptOS to POS vendors. In to inbound channel, CAC is doing direct sales..		
3. Anchor Chain	At mid-size chains: Win 50-200 stores deal trial (efficiency-fee) win. The totality of subscriptions and examples reference in the future dealing will include chain level, ROI guarantee.	1 anchor chain; 50–200 stores	Months 6–9
4.Scale	The inbound, reference case drives have the driver being the reference case driver. Chain HQs model (flattened platform fee; stores Per-Location SaaS). Vietnam playbook, launch to Thailand and Indonesia.	200 stores by Month 18; SEA entry	Months 9–18

6.3 REVENUE PROJECTIONS

They have also based their assumptions in projections on the basis of assuming modest assumptions of store base adoption that include; 10% potential available penetration of store base, morale store base of recovery and efficiency fee 40% of value of waste (average in the industry being 35-80%) and average LTV before churn 24 months.

Year	Stores	MRR (Month End)	ARR	Key Milestone
Year 1	20	$1,800	$180,000	3 chain pilots complete; KiotViet integration live
Year 2	200	$18,000	$1,800,000	First anchor chain converted; Series A preparation
Year 3	600	$54,000	$6,000,000	SEA expansion (TH/ID entry); Enterprise API revenue begins
Year 4	2,000	$150,000+	$18,000,000+	40% waste reduction proven at scale

The estimated SaaS subscriptions and efficiency fee flows of revenues are at hand. Enterprise API revenue which will be counted other than, monthly projections - considerable as an upside in Month 12.

7. EXECUTION ROADMAP

Phase 1: Foundation (Months 0–6)

Target: 3–5 stores, first recovered GMV proof point

- Build Virtual E-Ink (software shelf name MVP) is a prototype of a protocol named Web socket protocol.
- Deploy Agentic reasoning loop v1 -single dynamic pricing underlying loop only.
- Minimizes Human-in-the-Loop approvals dashboard to be investigated by manager.
- The new stores (3-5 pilot stores in the HCMC (District 1 and Phu My Hung).
- Realign with GDGoC 2026 competition as proof-of-concept.
- Measurable GMV rescue - demonstrate 30%+ waste reduction vs. baseline in pilot stores

Phase 2: Commercialisation (Months 6–12)

Target: 50 stores, $15,000 MRR

- Construct Procurement Already accessible (PO generation and routing) Build Autonomous Procurement Already accessible (PO generation and routing)
- Fuse macro signal layer Brightdata price feeds.
- Geo-pricing model Construct Geo-demographic (6 HCMC district) Items.
- Win and do first anchor chain deal 20-50 stores efficiency-fee trial.
- Signing agreement with KiotViet and/or MISA.
- Measuring infrastructure infrastructure Build Construct SaaS billing infrastructure.
- Target milestone: $15,000 MRR.

Phase 3: Scale (Months 12–18)

Target: 200 stores, $150,000 MRR

- Implement white-label pricing engine.
- Implement Enterprise API to POS regulars.
- Positive- negative Enter Thailand and Indonesia are sitting up with 2 anchor chains associates each.
- 200 store milestone It has the Series A narrative and values in position.
- To invent a carbon-credit portal, Scope 3 reporting software.
- E-Ink physical OEM contract of shelf labels.
- Target goal: $150,000 MRR